

Mayank Gandhi

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Education

Northeastern University, Boston, MA | *M.S. Bioinformatics* | **GPA: 3.9**

Sept 2024 – Present

Coursework: Bioinformatics Programming, Computational Biology, Data Analytics, Genomics, Transcriptomics

Maharashtra Institute of Technology, Pune, India | *B.Tech Bioengineering* | **GPA: 3.27**

Aug 2019 – Jun 2023

Coursework: Systems Biology, Molecular Biology, Pharmacology, Statistics, Medical Imaging, Nanotechnology

Skills

- **Software Engineering:** Python, R, Rust (rayon, extendr/PyO3 for cross-language performance modules), SQL, Bash/Unix, Git, Docker, Singularity, AWS (EC2, S3, IAM), unit testing, CI/CD, Angular/Spring Boot full-stack development
- **Pipeline & Systems Architecture:** Production-grade pipeline design and benchmarking frameworks (Nextflow, Snakemake), SLURM/HPC cluster computing, algorithm validation via synthetic ground-truth simulation, LLM-driven/agent analysis platforms, reproducible & version-controlled workflow engineering
- **Genomics & Transcriptomics:** Bulk & single-cell RNA-seq deconvolution and reference-atlas engineering (batch-aware integration, marker-based annotation), scRNA-seq/snRNA-seq (Seurat, Scanpy), spatial transcriptomics (Squidpy), single-cell foundation models (scGPT, Geneformer), multi-omics integration, differential expression, pathway enrichment (GSEA, GO ORA, Reactome), alignment/quantification (STAR, HISAT2, BWA, Salmon), variant calling (GATK), QC (Trimmomatic, FastQC), IGV
- **Statistical Modeling & ML:** Interpretable ML for biological readout prediction, supervised/unsupervised learning, ensemble methods, dimensionality reduction, model evaluation (AUC-ROC, F1, SHAP), Tableau, Power BI

Experience

Century Therapeutics, Inc., Boston, MA – Hybrid

Jun 2026 – Present

Computational Biology/Data Science, Quality Assurance Co-op

- Established a quantitative, evidence-based framework for selecting **bulk RNA-seq deconvolution** algorithms in cell-therapy product characterization, replacing ad-hoc method choice with systematic benchmarking of reference-based and reference-free approaches
- De-risked deconvolution accuracy claims by building a **ground-truth validation framework** using in-silico synthetic bulk mixtures at known cell-type proportions, giving the team a reusable standard for evaluating any future method
- Found that **reference resolution**, more than algorithm choice, drove deconvolution accuracy for transcriptionally similar subpopulations, and redirected pipeline investment toward single-cell reference atlas engineering (QC, batch-aware integration, marker-based annotation)
- Integrated multi-omics sequencing workflows into an internal **agent AI platform**, giving non-computational scientists automated, LLM-driven interpretation of high-dimensional expression data

Cencora (formerly AmerisourceBergen), Conshohocken, PA – Remote

Jun 2025 – Aug 2025

Software/EDI Summer Intern – Technical Delivery, D&S IT

- Designed and shipped a **Map Catalog UI** (Angular, Spring Boot) that consolidated fragmented **EDI transaction workflows** onto a single platform, backed by **Oracle SQL**-driven data mapping and automated QC reporting across X12-compliant healthcare data exchange operations

National Center for Biological Sciences (NCBS-TIFR), Bengaluru, India

Jan 2023 – Jul 2023

Computational Researcher & Modeler Intern

- Built and validated computational models of **dopaminergic signalling pathways** in Parkinson's disease (**HillTau**, ODE frameworks) that surfaced cell-type-specific synaptic dysfunction mechanisms; engineered reproducible multi-omics integration pipelines in Bash, validated against experimental data via **FindSim**

Projects

NeuroExprViz: Alzheimer's Transcriptomics | Northeastern University | GitHub | Live Dashboard

Feb 2026 – Mar 2026

- Engineered a differential expression & pathway enrichment pipeline (limma, clusterProfiler) across **GSE5281** (n=161, 6 brain regions), surfacing **3,358 DEGs** and **27 Hallmark pathways** implicating mitochondrial and neuroinflammatory dysregulation in AD progression
- Shipped an interactive, publicly deployed **monitoring dashboard** (R, Python, Plotly.js, GitHub Pages) that lets non-computational collaborators explore region-level expression trends without touching raw data

CellChatR: Rust Rewrite of CellChat | Personal Project | GitHub

Jul 2026 – Sept 2026

- Rebuilt the core cell-cell communication inference engine of **CellChat** (R) in **Rust**, cutting compute overhead by parallelizing signalling-pathway computations with **rayon**
- Preserved full backward compatibility with existing R workflows by exposing the Rust engine through **extendr** bindings, making the performance gain a drop-in upgrade; package submitted to **Bioconductor**

COPD Transcriptomic Analysis | Northeastern University | GitHub

Jan 2026 – Mar 2026

- Built a statistical workflow (limma, clusterProfiler, fgsea) across **GSE76925** (111 COPD vs 40 controls) that flagged biologically significant process signals (ECM remodelling, ribosomal suppression) missed by standard differential expression alone

Leadership

Graduate Student Ambassador, College of Science, Northeastern University, Boston, MA

Jul 2026 – Present

- Represented the College of Science at admitted-student and recruiting events; mentored and advised prospective and incoming graduate students on program navigation and academic planning